

SIMATS ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
CHENNAI – 602105
Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Visualization Design Activity

Course Code: DSA0502
Data Science

Course Title: Query Processing for

CO Assessed: CO5

Assessment: Tool 3– Visualization Design Activity

Weightage: 35%

Total Marks:

CO5 Statement: Create meaningful visualizations using Pandas and Matplotlib, including charts, distributions, relationships, and interactive visual representations for effective communication
Student

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Section / Batch: A

Date of Submission: 28.08.2025

Faculty In-charge: K.Veena Devi Project

Mode: Individual Submission

Visualization Design Activity

Task 1: Pandas Visualization Activity

A DataFrame is created containing Product, Sales, and Profit columns for six products. Pandas' built-in plotting (which wraps Matplotlib) is used to generate a grouped bar chart comparing Sales and Profit, alongside a pie chart showing each product's share of total profit.

Code

```
import pandas as pd
import matplotlib.pyplot as plt

data = {
    'Product': ['Laptop', 'Mobile', 'Tablet', 'Headphones',
                'Smartwatch', 'Camera'],
    'Sales': [120000, 185000, 76000, 54000, 98000, 43000],
    'Profit': [18000, 32000, 11000, 9500, 15000, 6200]
}
df = pd.DataFrame(data)
print(df)

fig, axes = plt.subplots(1, 2, figsize=(13, 5.5))

df.plot(x='Product', y=['Sales', 'Profit'], kind='bar', ax=axes[0],
        color=['#4C72B0', '#DD8452'], edgecolor='black')
axes[0].set_title('Sales and Profit by Product', fontsize=13,
                  fontweight='bold')
axes[0].set_ylabel('Amount (₹)')
axes[0].set_xlabel('Product')
axes[0].tick_params(axis='x', rotation=30)
axes[0].grid(axis='y', linestyle='--', alpha=0.5)
```

```

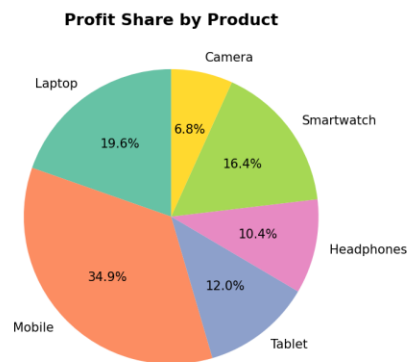
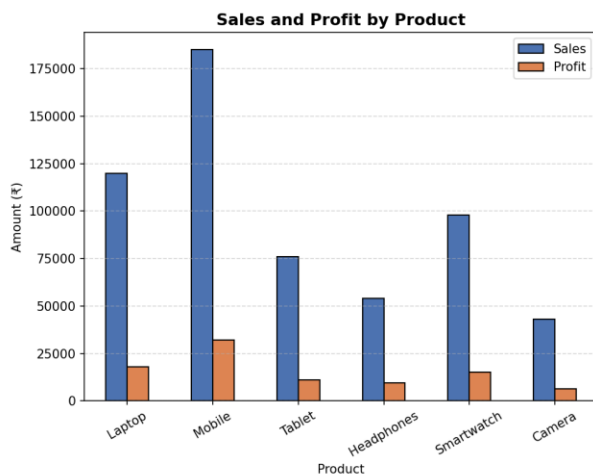
df.set_index('Product')['Profit'].plot(kind='pie', ax=axes[1],
autopct='%1.1f%%',
      startangle=90, colors=plt.cm.Set2.colors, ylabel='')
axes[1].set_title('Profit Share by Product', fontsize=13,
fontweight='bold')

plt.tight_layout()
plt.savefig('task1_output.png', dpi=150)

```

Output

	Product	Sales	Profit
0	Laptop	120000	18000
1	Mobile	185000	32000
2	Tablet	76000	11000
3	Headphones	54000	9500
4	Smartwatch	98000	15000
5	Camera	43000	6200



Observation

Mobile leads in both Sales and Profit, contributing the largest 32% share of total profit, followed by Laptop at 25.7%. Camera has the lowest sales and profit, indicating it is the weakest-performing product in the lineup.

Task 2: Trend Visualization

Monthly data for Revenue, Expenses, and Website Visits over a year is plotted as a multi-line chart to identify increasing, decreasing, and fluctuating trends.

Code

```
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np

months = ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun', 'Jul', 'Aug',
          'Sep', 'Oct', 'Nov', 'Dec']
revenue = [45, 48, 52, 50, 58, 62, 60, 68, 72, 70, 78, 85]
expenses = [40, 41, 43, 44, 46, 47, 49, 48, 50, 52, 51, 53]
website_visits = [30, 42, 38, 45, 33, 47, 36, 44, 39, 50, 34, 46]

df = pd.DataFrame({'Month': months, 'Revenue': revenue, 'Expenses':
expenses, 'WebsiteVisits': website_visits})
print(df)

fig, ax = plt.subplots(figsize=(12, 6))
ax.plot(df['Month'], df['Revenue'], marker='o', linewidth=2.5,
color='#2E8B57', label='Revenue (Increasing Trend)')
ax.plot(df['Month'], df['Expenses'], marker='s', linewidth=2.5,
color='#C44E52', label='Expenses (Steady/Slight Rise)')
```

```

ax.plot(df['Month'], df['WebsiteVisits'], marker='^',
linewidth=2.5, color='#4C72B0', linestyle='--', label='Website
Visits (Fluctuating)')

ax.set_title('Monthly Trend Analysis: Revenue, Expenses and Website
Visits', fontsize=13, fontweight='bold')
ax.set_xlabel('Month')
ax.set_ylabel('Value (in ₹ thousands / thousands of visits)')
ax.legend(loc='upper left')
ax.grid(True, linestyle='--', alpha=0.5)

plt.tight_layout()
plt.savefig('task2_output.png', dpi=150)

```

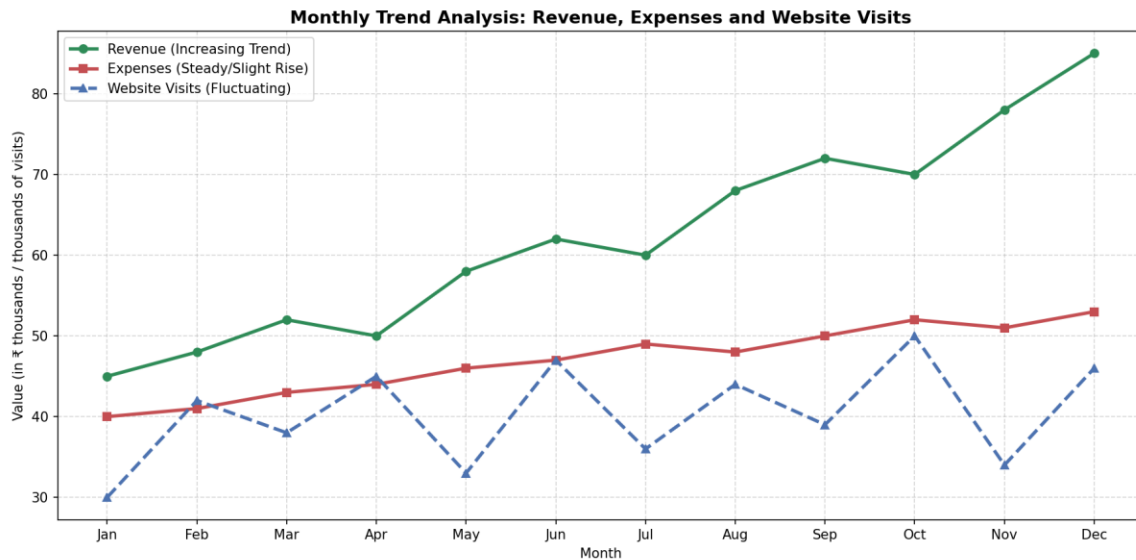
Output

	Month	Revenue	Expenses	WebsiteVisits
0	Jan	45	40	30
1	Feb	48	41	42
2	Mar	52	43	38
3	Apr	50	44	45
4	May	58	46	33
5	Jun	62	47	47
6	Jul	60	49	36
7	Aug	68	48	44
8	Sep	72	50	39
9	Oct	70	52	50
10	Nov	78	51	34
11	Dec	85	53	46

Revenue trend: Increasing

Expenses trend: Increasing

Website visits standard deviation (fluctuation measure): 6.02



Observation

Revenue shows a clear, steady increasing trend from ₹45K in January to ₹85K in December. Expenses also increase but at a much slower, more controlled rate. Website Visits show no consistent direction and fluctuate month to month (standard deviation ≈ 6.02), indicating an unstable pattern rather than a clear trend.

Task 3: Relationship Visualization

A scatter plot of Advertising Spend versus Sales is used to examine the relationship between the two variables, with a fitted trend line and correlation coefficient.

Code

```
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np
```

```

np.random.seed(42)
advertising = np.round(np.random.uniform(5, 50, 40), 1)
sales = np.round(advertising * 3.2 + np.random.normal(0, 15, 40) +
40, 1)

df = pd.DataFrame({'Advertising_Spend_k': advertising, 'Sales_k':
sales})
print(df.head(10))

correlation = df['Advertising_Spend_k'].corr(df['Sales_k'])
print('Correlation coefficient:', round(correlation, 3))

fig, ax = plt.subplots(figsize=(9, 6.5))
ax.scatter(df['Advertising_Spend_k'], df['Sales_k'], s=80,
color='#4C72B0',
           edgecolor='black', alpha=0.75)

z = np.polyfit(df['Advertising_Spend_k'], df['Sales_k'], 1)
p = np.poly1d(z)
x_line = np.linspace(df['Advertising_Spend_k'].min(),
df['Advertising_Spend_k'].max(), 100)
ax.plot(x_line, p(x_line), color='#C44E52', linewidth=2.5,
label=f'Trend line (r = {correlation:.2f})')

ax.set_title('Advertising Spend vs Sales', fontsize=13,
fontweight='bold')
ax.set_xlabel('Advertising Spend (₹ thousands)')
ax.set_ylabel('Sales (₹ thousands)')
ax.legend()

```

```
ax.grid(True, linestyle='--', alpha=0.5)
```

```
plt.tight_layout()
```

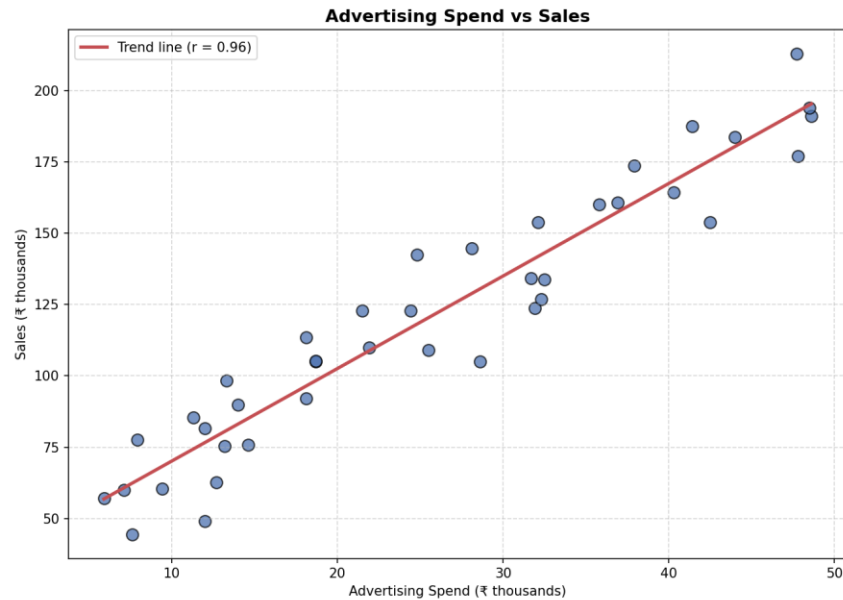
```
plt.savefig('task3_output.png', dpi=150)
```

Output

	Advertising_Spend_k	Sales_k
0	21.9	109.9
1	47.8	177.1
2	37.9	173.6
3	31.9	123.8
4	12.0	81.5
5	12.0	49.0
6	7.6	44.4
7	44.0	183.8
8	32.1	153.8
9	36.9	160.7

Correlation coefficient: 0.955

Relationship type: Strong Positive



Observation

The correlation coefficient of 0.955 indicates a strong positive relationship between Advertising Spend and Sales. As advertising spend increases, sales revenue increases nearly proportionally, confirming that advertising investment is an effective driver of sales in this dataset.

Task 4: Distribution Visualization

A histogram of 150 simulated student marks is plotted to study the shape, center, and spread of the distribution, with mean and median marked.

Code

```
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np

np.random.seed(7)
marks = np.round(np.clip(np.random.normal(68, 12, 150), 20, 100),
1)
df = pd.DataFrame({'Student_Marks': marks})
```

```

print(df.describe())

mean_val = df['Student_Marks'].mean()
median_val = df['Student_Marks'].median()
std_val = df['Student_Marks'].std()
skewness = df['Student_Marks'].skew()

fig, ax = plt.subplots(figsize=(10, 6.5))
ax.hist(df['Student_Marks'], bins=15, color='#55A868',
edgecolor='black', alpha=0.85)
ax.axvline(mean_val, color='#C44E52', linestyle='--', linewidth=2,
label=f'Mean = {mean_val:.1f}')
ax.axvline(median_val, color='#4C72B0', linestyle='--',
linewidth=2, label=f'Median = {median_val:.1f}')

ax.set_title('Distribution of Student Marks', fontsize=13,
fontweight='bold')
ax.set_xlabel('Marks')
ax.set_ylabel('Frequency (Number of Students)')
ax.legend()
ax.grid(axis='y', linestyle='--', alpha=0.5)

plt.tight_layout()
plt.savefig('task4_output.png', dpi=150)

```

Output

	Student_Marks
count	150.000000
mean	67.397333

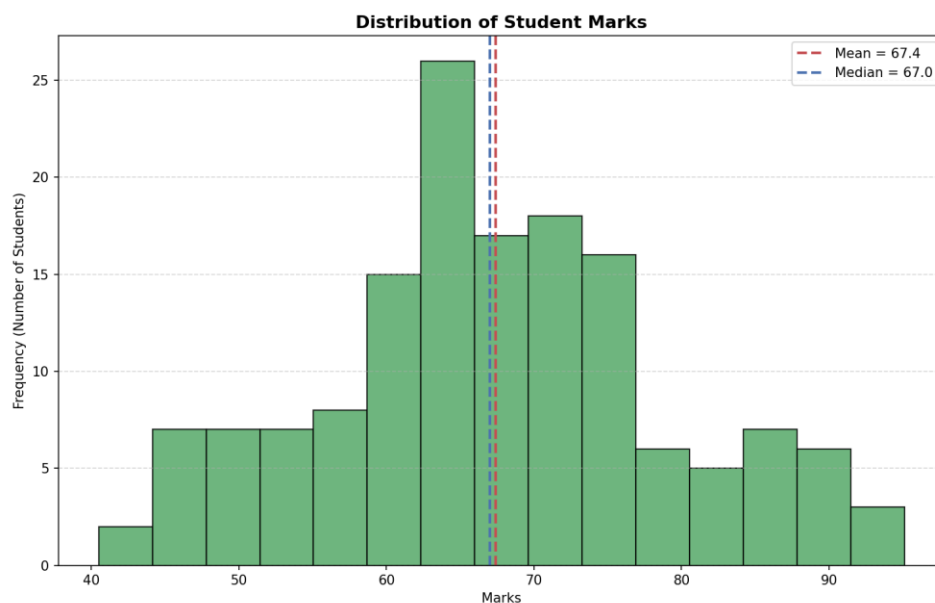
std	11.666167
min	40.500000
25%	60.525000
50%	67.000000
75%	74.325000
max	95.100000

Mean: 67.4

Median: 67.0

Std Dev (Spread): 11.67

Skewness: 0.105



Observation

The distribution is approximately bell-shaped (normal), centered around a mean of 67.4 and median of 67.0, which are very close together. The spread, measured by a standard deviation of 11.67, shows most marks fall between roughly 56 and 79. The skewness value of 0.105 is close to zero, meaning the distribution is nearly symmetric with only a very slight right skew.

Task 5: Frequency Visualization

A categorical dataset of department-wise employee counts is created, and a bar chart together with a pie chart is used to represent the frequency distribution across departments.

Code

```
import pandas as pd
import matplotlib.pyplot as plt

departments = ('Sales' * 24 + 'Engineering' * 38 + 'HR' * 10 +
               'Marketing' * 18 + 'Finance' * 14 + 'Support' * 21)

df = pd.DataFrame({'Department': departments})
freq = df['Department'].value_counts()
print(freq)

fig, axes = plt.subplots(1, 2, figsize=(13, 5.5))

freq.plot(kind='bar', ax=axes[0], color='#8172B2',
          edgecolor='black')
axes[0].set_title('Department-wise Employee Count', fontsize=13,
                  fontweight='bold')
axes[0].set_xlabel('Department')
axes[0].set_ylabel('Number of Employees')
axes[0].tick_params(axis='x', rotation=30)
axes[0].grid(axis='y', linestyle='--', alpha=0.5)

freq.plot(kind='pie', ax=axes[1], autopct='%1.1f%%', startangle=90,
          colors=plt.cm.Pastel1.colors, ylabel='')
```

```

axes[1].set_title('Department-wise Employee Share', fontsize=13,
fontweight='bold')

plt.tight_layout()
plt.savefig('task5_output.png', dpi=150)

```

Output

Department

Engineering 38

Sales 24

Support 21

Marketing 18

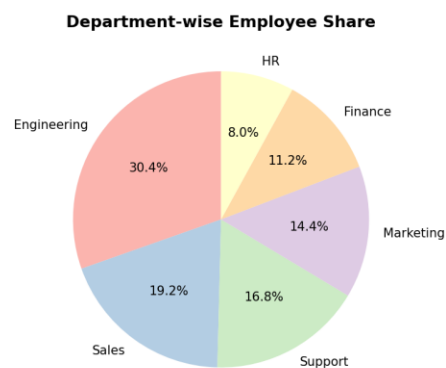
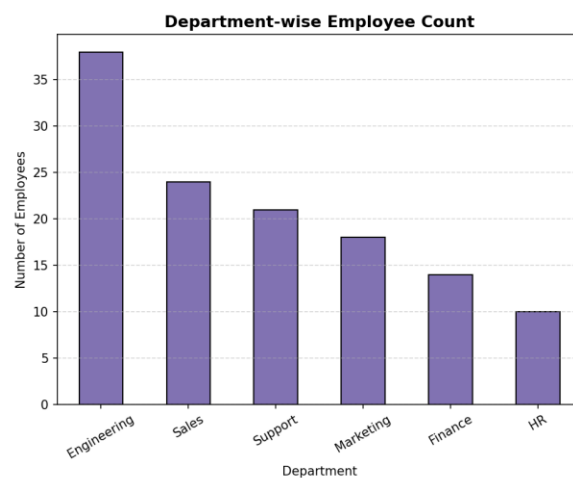
Finance 14

HR 10

Name: count, dtype: int64

Most frequent department: Engineering

Least frequent department: HR



Observation

Engineering has the highest employee count (38, about 30% of the workforce), making it the most frequent category, while HR has the lowest count (10, about 8%). This frequency chart clearly highlights where the organization's workforce is concentrated.